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THE COMPLETE GUIDE

Java 8

Lambdas, Streams & the Functional Revolution

A complete, example-driven guide to Java 8 — the release that changed Java forever — covering lambdas, functional interfaces, the Streams API, collectors, Optional, and the new Date/Time API.

24	130+	13	6
Chapters	Code Examples	Diagrams	Modules

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Java 8: The Complete Guide

First Edition · 2026

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MODULE 1 · FUNCTIONAL FOUNDATIONS

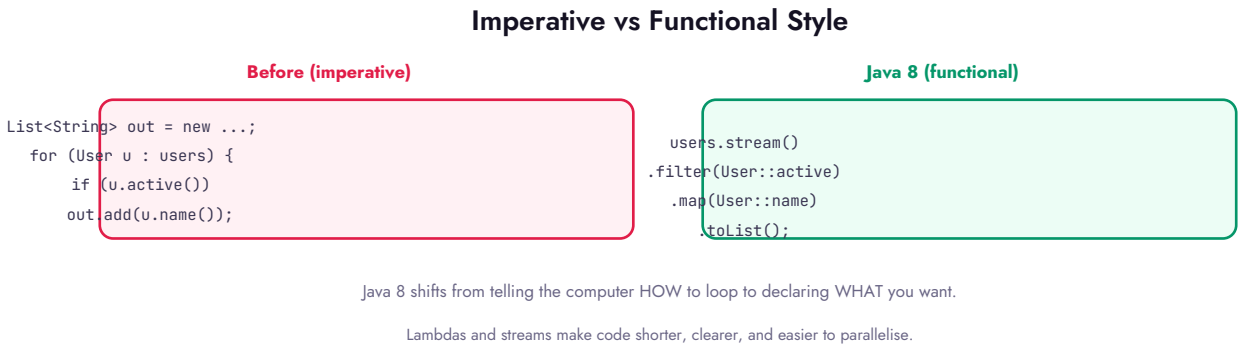
CHAPTER 01

Why Java 8 Changed Everything

The shift to a functional style, and why Java 8 remains the most important release.

Java 8, released in 2014, is the most significant release in the language's history. It introduced **lambdas** and **streams**, bringing a functional style to Java and changing how millions of developers write code every day.

Imperative to functional



Java 8 shifts from telling the computer HOW to loop to declaring WHAT you want.

The old imperative style spells out every step: create a list, loop, test a condition, add to the list. The functional style declares the intent — filter the active users, take their names — leaving the how to the library.

What Java 8 introduced

- **Lambda expressions** — concise anonymous functions.
- **Streams** — declarative processing of collections.
- **Functional interfaces** and **method references**.
- **Default methods** — evolving interfaces without breaking code.
- **Optional** — a better way to handle absent values.
- **A new Date/Time API** — replacing the broken old one.

Why it still matters

Every Java version since builds on these foundations, and Java 8 remains one of the most widely used releases in production. Mastering it is essential — for interviews, for reading real code, and for writing idiomatic modern Java.

> **The functional turning point**
> Java 8 brought lambdas, streams, functional interfaces, Optional, and a new Date/Time API, shifting Java toward a declarative, functional style. It is the foundation every later version builds on — and essential to master.

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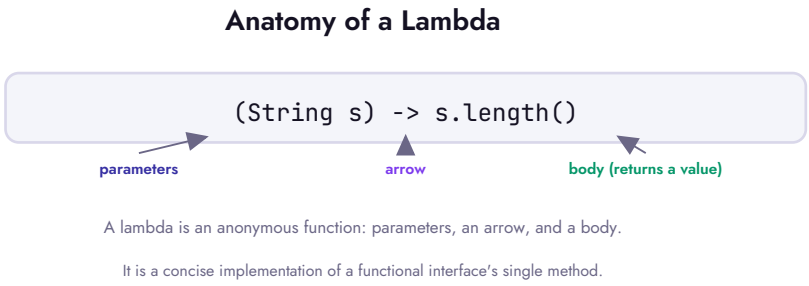
CHAPTER 02

Lambda Expressions

Anonymous functions that make behaviour something you can pass around.

A **lambda expression** is an anonymous function — a block of behaviour you can pass as an argument, return from a method, or store in a variable. Lambdas are the core feature that makes everything else in Java 8 possible.

The syntax



A lambda is parameters, an arrow, and a body — a concise implementation of a single method.

java

```
// (parameters) -> body
(String s) -> s.length()
s -> s.length()                 // type inferred, one param
(a, b) -> a + b                 // two params
() -> System.out.println("hi")  // no params
s -> { return s.trim(); }       // block body
```

Replacing anonymous classes

Before lambdas, passing behaviour meant a verbose anonymous class. A lambda expresses the same thing in one line — the compiler figures out the rest from the target type.

java

```
// before
button.addActionListener(new ActionListener() {
    public void actionPerformed(ActionEvent e) { save(); }
});
// Java 8
button.addActionListener(e -> save());
```

Capturing variables

A lambda can use variables from its surrounding scope, but they must be **effectively final** — not reassigned after the lambda captures them. This keeps lambdas safe to run later, possibly on another thread.

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Behaviour as a value

A lambda is an anonymous function — parameters, an arrow, a body — that you can pass around like data. It replaces verbose anonymous classes and can capture effectively-final variables from its scope.

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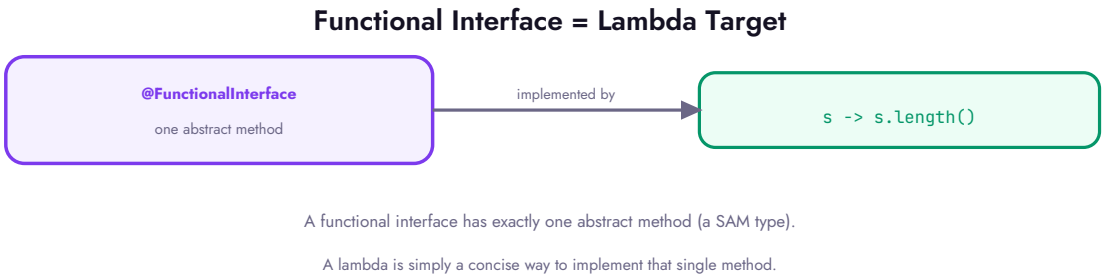
CHAPTER 03

Functional Interfaces

The single-method interfaces that give lambdas their type.

A lambda needs a type, and that type is a **functional interface** — an interface with exactly one abstract method. The lambda provides the implementation of that method.

One abstract method



A functional interface has one abstract method; a lambda implements it concisely.

```java
@FunctionalInterface
interface Transformer {
    String apply(String input);   // the single abstract method
}

Transformer upper = s -> s.toUpperCase();
upper.apply("hello");   // "HELLO"
```

The @FunctionalInterface annotation

Marking an interface `@FunctionalInterface` is optional but recommended — the compiler then enforces that it has exactly one abstract method, catching mistakes early. Default and static methods do not count against this.

Why this design

Because a lambda is just an implementation of a single-method interface, lambdas integrate seamlessly with existing Java — any old interface with one method (like `Runnable` or `Comparator`) instantly accepts a lambda, with no library changes.

> **TIP**
> You rarely need to write your own functional interfaces — the JDK ships a rich set (`Predicate`, `Function`, `Consumer`, `Supplier`) covered in the next chapters.

The type behind a lambda

A functional interface has exactly one abstract method, and a lambda implements it. The @FunctionalInterface annotation enforces this. Any single-method interface — old or new — accepts a lambda automatically.

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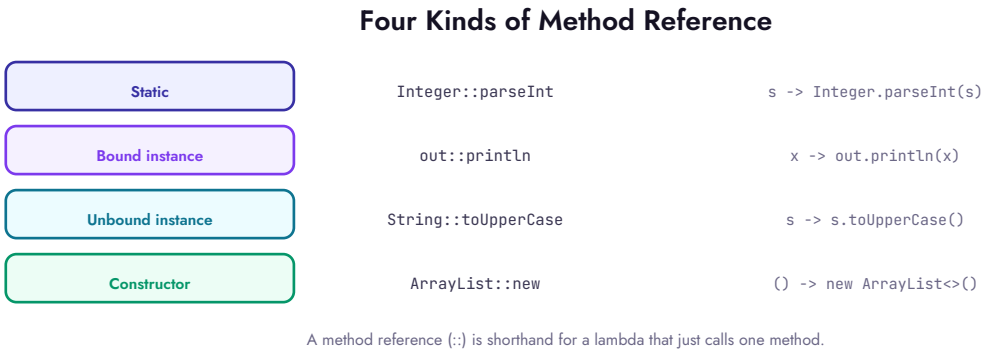
CHAPTER 04

Method References

An even shorter syntax for lambdas that just call an existing method.

When a lambda does nothing but call one existing method, a **method reference** expresses it more concisely. `s -> s.toUpperCase()` becomes `String::toUpperCase`.

The four kinds



A method reference (::) is shorthand for a lambda that just calls one method.

Kind	Example
Static method	Integer::parseInt
Instance of an object	System.out::println
Instance of a type	String::toUpperCase
Constructor	ArrayList::new

When to use them

Reach for a method reference when the lambda is a plain pass-through to a named method — it reads better. Keep a lambda when there is extra logic, or when a method reference would obscure what is happening.

```java
names.stream().map(String::toUpperCase).toList();
numbers.stream().map(Integer::parseInt).toList();
users.stream().map(User::new).toList();   // constructor ref
```

The unbound instance reference

The trickiest kind is `String::toUpperCase` — the method is called on the first lambda argument. So `String::toUpperCase` means `s -> s.toUpperCase()`, with the stream element becoming the receiver.

Shorthand for a single call

A method reference (Type::method) is a compact lambda that just calls one method — static, on a specific object, on the argument's type, or a constructor. Use it when it reads more clearly than the lambda.

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MODULE 2 · INTERFACES & BUILDING BLOCKS

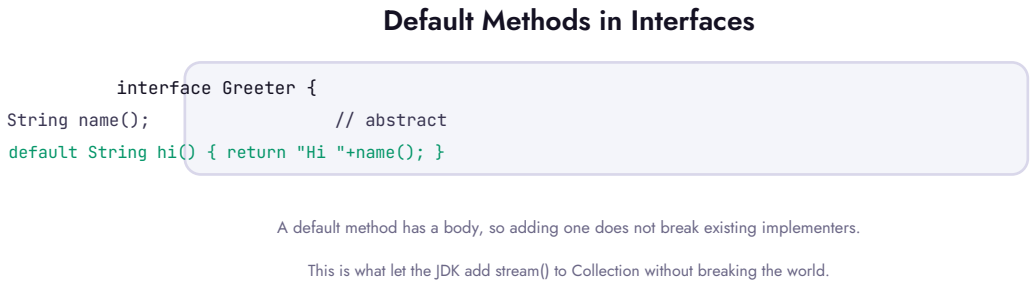
CHAPTER 05

Default & Static Methods

Adding method bodies to interfaces so they can evolve without breaking code.

Before Java 8, adding a method to an interface broke every class that implemented it. **Default methods** — interface methods with a body — solved this, and are the reason the JDK could add `stream()` to every collection.

A method with a body



A default method has a body, so adding one does not break existing implementers.

```java
interface Greeter {
    String name();                      // abstract
    default String greet() {            // default: has a body
        return "Hello, " + name();
    }
}
// implementers get greet() for free, can override it
```

Why they exist

Default methods let an interface gain new behaviour without forcing every implementer to change. This backward-compatible evolution is exactly how `Collection.stream()`, `Iterable.forEach()`, and `Comparator` helpers were added in Java 8.

Static interface methods

Interfaces can also have **static** methods — utility methods that belong to the interface itself, like factory or helper functions. `Comparator.comparing(...)` is a static interface method.

The diamond question

WATCH OUT

If a class inherits the same default method from two interfaces, it must override it to resolve the conflict — the compiler will not guess. This keeps multiple-inheritance-of-behaviour explicit and safe.

Interfaces that carry behaviour

Default methods give interface methods a body, letting interfaces evolve without breaking implementers — the mechanism behind stream() and forEach(). Static interface methods provide related utilities like Comparator.comparing.

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MODULE 2 · INTERFACES & BUILDING BLOCKS

CHAPTER 06

The Built-in Functional Interfaces

The core interfaces in java.util.function that you will use constantly.

You rarely need to define your own functional interface — the `java.util.function` package provides a small set that covers almost every case. Learn these four and you know most of them.

The core four

Interface	Method	Purpose
Predicate<T>	test(T) -> boolean	a condition
Function<T,R>	apply(T) -> R	a transformation
Consumer<T>	accept(T) -> void	an action
Supplier<T>	get() -> T	a producer

java

```
Predicate<String> isEmpty = s -> s.isEmpty();
Function<String,Integer> length = s -> s.length();
Consumer<String> print = s -> System.out.println(s);
Supplier<LocalDate> today = LocalDate::now;
```

The variants

- **BiFunction, BiPredicate, BiConsumer** — two-argument versions.
- **UnaryOperator, BinaryOperator** — same input and output type.
- **IntPredicate, ToIntFunction**, etc. — primitive specialisations that avoid boxing.

Where you meet them

These interfaces are the parameter types throughout the streams API: `filter` takes a `Predicate`, `map` takes a `Function`, `forEach` takes a `Consumer`. Recognising them makes the whole API click.

Four interfaces cover most cases

Predicate (test), Function (transform), Consumer (act), and Supplier (produce) are the core functional interfaces, with Bi-, operator, and primitive variants. They are the parameter types across the streams API.

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MODULE 2 · INTERFACES & BUILDING BLOCKS

CHAPTER 07

Composing Functions

Building complex behaviour by chaining small functions together.

The functional interfaces come with **default methods** that let you combine them. Small, reusable functions compose into larger behaviour — a hallmark of functional programming.

Chaining functions

`Function` has `andThen` (run this, then that) and `compose` (run that first). They build a pipeline of transformations from small pieces.

java

```
Function<Integer,Integer> times2 = x -> x * 2;
Function<Integer,Integer> plus1 = x -> x + 1;

times2.andThen(plus1).apply(5);   // (5*2)+1 = 11
times2.compose(plus1).apply(5);   // (5+1)*2 = 12
```

Combining predicates

`Predicate` offers `and`, `or`, and `negate` to build compound conditions from simple ones — more readable than a tangle of boolean operators.

java

```
Predicate<String> notEmpty = s -> !s.isEmpty();
Predicate<String> shortEnough = s -> s.length() < 20;
Predicate<String> valid = notEmpty.and(shortEnough);
```

Why compose

- **Reuse** — define a check or transform once, combine many ways.
- **Readability** — named pieces beat inline boolean logic.
- **Testability** — test each small function on its own.

Small functions, combined

Functional interfaces compose via default methods: Function.andThen/compose chain transformations, and Predicate.and/or/negate build compound conditions. Composing small, named, testable pieces is the functional way.

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MODULE 2 · INTERFACES & BUILDING BLOCKS

CHAPTER 08

Comparators the Functional Way

Building readable sorting logic with the new Comparator helpers.

Sorting used to mean verbose `Comparator` classes with hand-written comparison logic. Java 8's `Comparator` helpers make sorting expressive, readable, and easy to chain.

comparing

`Comparator.comparing` builds a comparator from a key extractor — a function that pulls the value to sort by. No more manual `compareTo` plumbing.

```java
users.sort(Comparator.comparing(User::name));
// sort by name, ascending
```

Chaining and reversing

`thenComparing` adds tie-breakers, and `reversed` flips the order — expressing multi-key sorts that read like a sentence.

```java
users.sort(
    Comparator.comparing(User::city)
              .thenComparing(User::name)
              .reversed());
```

Handling nulls and primitives

Helper	Does
comparing	sort by an extracted key
thenComparing	add a tie-breaker
reversed	reverse the order
nullsFirst / nullsLast	handle null keys safely
comparingInt	avoid boxing for int keys

Readable, chainable sorting

Comparator.comparing builds a comparator from a key extractor; thenComparing adds tie-breakers and reversed flips order. With nullsFirst and comparingInt, sorting becomes expressive and safe.

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MODULE 3 · THE STREAMS API

CHAPTER 09

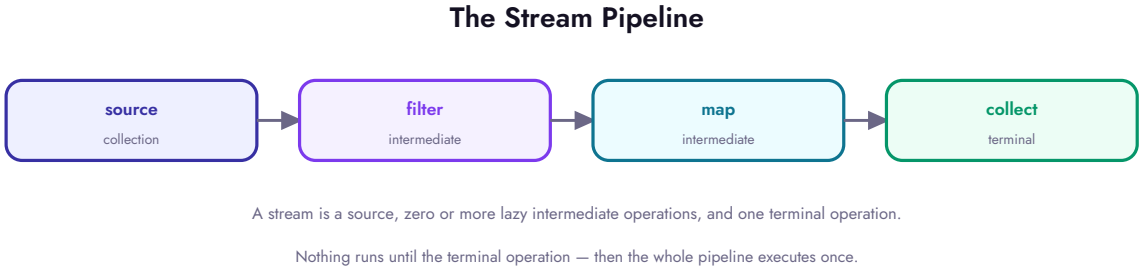
Introduction to Streams

What a stream is, how a pipeline works, and why streams are lazy.

09

The **Streams API** is Java 8's flagship feature — a declarative way to process sequences of data. A stream is not a collection; it is a pipeline of operations over a source.

The pipeline



A stream is a source, zero or more lazy intermediate operations, and one terminal operation.

```java
List<String> names = users.stream()   // source
    .filter(User::isActive)           // intermediate
    .map(User::name)                  // intermediate
    .sorted()                         // intermediate
    .toList();                        // terminal
```

What makes streams different

- **Not a data structure** — a stream carries no storage; it processes a source.
- **Lazy** — intermediate operations do nothing until a terminal operation runs.
- **Functional** — they do not mutate the source; they produce a result.
- **Single-use** — a stream is consumed once; reuse throws an exception.

Laziness in action

Intermediate operations just build up the pipeline. Nothing runs until the terminal operation is invoked — then the whole chain executes in a single pass, which lets the library optimise (for example, stopping early on `findFirst`).

A lazy pipeline, not a collection

A stream is a pipeline: a source, lazy intermediate operations, and one terminal operation that triggers execution. Streams are functional, single-use, and process data in one optimised pass — they are not storage.

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MODULE 3 · THE STREAMS API

CHAPTER 10

Creating Streams

The many ways to obtain a stream from your data.

Before you can process a stream, you need one. Streams come from collections, arrays, individual values, ranges, and generators — covering every source you are likely to have.

The common sources

```java
list.stream();                      // from a Collection
Arrays.stream(array);               // from an array
Stream.of("a", "b", "c");            // from values
IntStream.range(0, 10);             // a range of ints
"line1\nline2".lines();             // lines of text
Stream.iterate(1, n -> n * 2).limit(10);   // generated
```

Primitive streams

`IntStream`, `LongStream`, and `DoubleStream` avoid the cost of boxing and add numeric operations like `sum()`, `average()`, and `range()`. Use them for number-heavy work.

```java
int total = IntStream.rangeClosed(1, 100).sum();   // 5050
OptionalDouble avg = scores.stream().mapToInt(Integer::intValue).average();
```

Infinite streams

`Stream.iterate` and `Stream.generate` produce **infinite** streams — you must bound them with `limit` (or a predicate), or the pipeline never ends.

WATCH OUT

An infinite stream without a limiting operation will run forever. Always pair `generate/iterate` with `limit` or a short-circuiting terminal operation.

Streams from anything

Obtain streams from collections (stream()), arrays (Arrays.stream), values (Stream.of), ranges (IntStream.range), and generators (iterate/generate, which need a limit). Primitive streams avoid boxing for numeric work.

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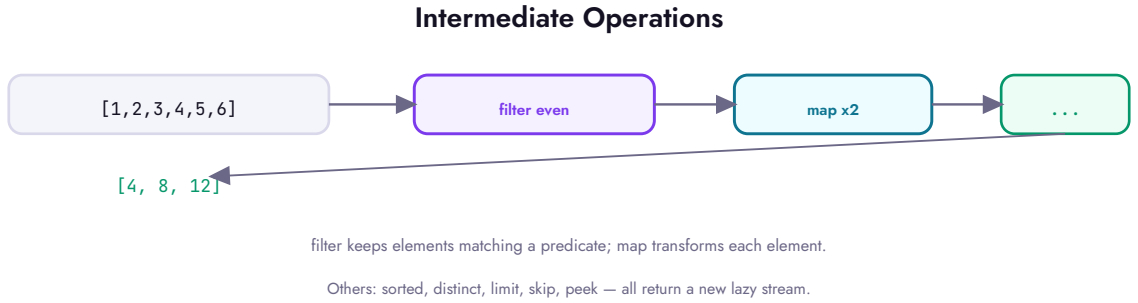
CHAPTER 11

Intermediate Operations

The lazy operations that transform a stream into another stream.

Intermediate operations transform a stream and return a new stream, so they chain. They are lazy — building up the pipeline without doing work until a terminal operation runs.

filter and map



filter keeps elements matching a predicate; map transforms each element.

`filter` keeps only elements matching a predicate; `map` transforms each element into another. These two do most of the everyday work.

```java
orders.stream()
    .filter(o -> o.total() > 100)    // keep large orders
    .map(Order::customer)            // to their customers
    .distinct()                      // unique
    .toList();
```

The full toolkit

Operation	Does
filter	keep elements matching a predicate
map	transform each element
sorted	sort the elements
distinct	remove duplicates
limit / skip	take or drop the first n
peek	observe elements (for debugging)

Laziness matters

Because they are lazy, intermediate operations are fused: with `limit` after a `map`, only as many elements as needed are mapped. The library processes each element through the whole chain rather than materialising intermediate lists.

Lazily transform the stream

Intermediate operations (filter, map, sorted, distinct, limit, skip, peek) return a new lazy stream and chain together. filter and map do most of the work; laziness lets the pipeline process only what is needed.

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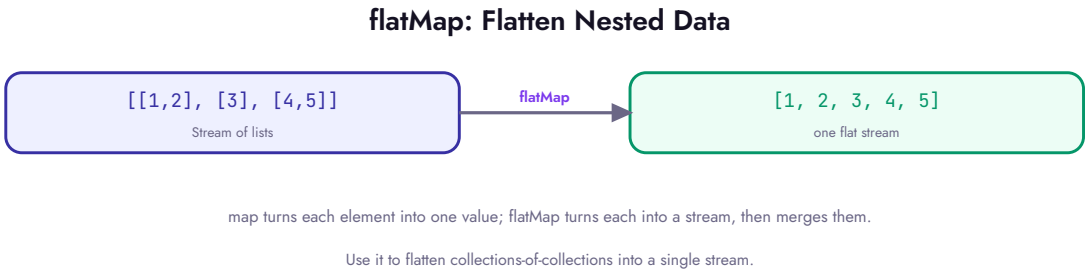
CHAPTER 12

flatMap & Flattening

Turning a stream of collections into a single flat stream.

`map` turns each element into exactly one value. But sometimes each element yields many values — a user has many orders, a line has many words. `flatMap` handles this by flattening.

One-to-many, then flatten



map turns each element into one value; flatMap turns each into a stream, then merges them.

`flatMap` takes a function that turns each element into a stream, then concatenates all those streams into one. A `Stream<List<X>>` becomes a `Stream<X>`.

```java
// each user has a List<Order>; get all orders as one stream
List<Order> all = users.stream()
    .flatMap(u -> u.orders().stream())
    .toList();
```

map vs flatMap

	map	flatMap
Function returns	one value	a stream of values
Result	same count	flattened, merged
Use for	1-to-1 transform	1-to-many, nesting

A common use: splitting

```java
// all words across many lines
lines.stream()
    .flatMap(line -> Arrays.stream(line.split(" ")))
    .toList();
```

Flatten one-to-many

flatMap maps each element to a stream and merges them into one — turning a Stream of collections into a single flat stream. Use it whenever each element expands into many, like users-to-orders or lines-to-words.

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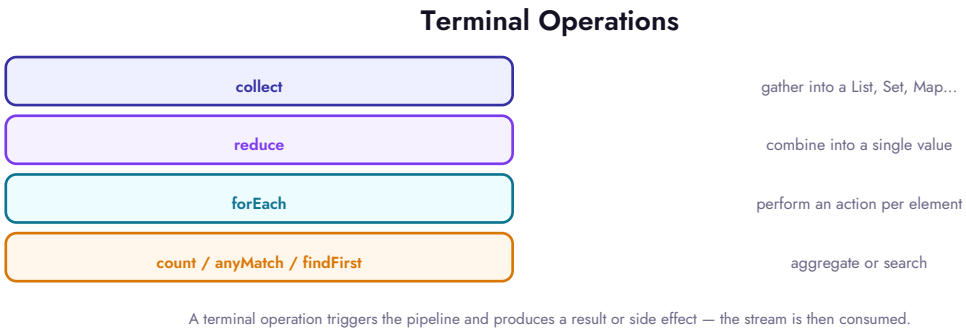
CHAPTER 13

Terminal Operations

The operations that trigger the pipeline and produce a result.

Intermediate operations build the pipeline, but nothing happens until a **terminal operation** runs it. The terminal operation produces a result — a value, a collection, or a side effect — and consumes the stream.

The main terminals



A terminal operation triggers the pipeline and produces a result or side effect.

Operation	Produces
collect	a List, Set, Map, or String
reduce	a single combined value
forEach	a side effect per element
count	the number of elements
anyMatch / allMatch	a boolean test
findFirst / findAny	an Optional element

Collecting results

```java
List<String> names = users.stream().map(User::name).toList();
long active = users.stream().filter(User::isActive).count();
Optional<User> first = users.stream().findFirst();
```

Short-circuiting

Some terminals stop early. `findFirst`, `anyMatch`, and `limit` can finish without processing the whole stream — which is why laziness matters and why they work even on infinite streams.

WATCH OUT

A stream can be consumed only once. After a terminal operation, the stream is spent — calling another operation on it throws `IllegalStateException`. Create a fresh stream for each pipeline.

Trigger the pipeline, get a result

Terminal operations (collect, reduce, forEach, count, match, find) run the pipeline and produce a result, consuming the stream. Some short-circuit, finishing early — and a stream can only be used once.

MODULE 3 · THE STREAMS API

CHAPTER 14

reduce & Aggregation

Combining a stream's elements into a single result.

`reduce` is the most general terminal operation — it combines all elements into one result by repeatedly applying a binary operation. Sum, product, concatenation, maximum: all are reductions.

The forms of reduce

```java
// with an identity and an accumulator
int sum = nums.stream().reduce(0, (a, b) -> a + b);

// without identity -> returns an Optional
Optional<Integer> max = nums.stream().reduce(Integer::max);

// sum via a built-in
int total = nums.stream().mapToInt(Integer::intValue).sum();
```

How it works

`reduce` starts from an identity value and folds each element in with the accumulator function — `((0+a)+b)+c`. The identity is the starting point and the result when the stream is empty.

Prefer specialised operations

For common aggregations, prefer the purpose-built operations — `sum()`, `average()`, `count()`, `max()` on primitive streams, or the summarising collectors — they are clearer than a hand-written `reduce`.

The identity must be neutral

WATCH OUT

The identity value must be neutral for the operation (0 for sum, 1 for product, "" for concatenation). A wrong identity silently produces wrong results, especially in parallel.

Fold many into one

reduce combines all elements into a single value using an identity and an accumulator; without an identity it returns an Optional. Prefer built-ins like sum() and the summarising collectors for common aggregations.

MODULE 4 · COLLECTORS & PARALLELISM

CHAPTER 15

Collectors: Gathering Results

Turning a stream into lists, sets, maps, and strings with collect.

The `collect` terminal operation, paired with a **Collector**, is the most versatile way to gather a stream's results — into collections, maps, joined strings, and aggregated values.

The common collectors



collect() uses a Collector to turn a stream into a list, set, map, string, or grouped structure.

```java
import static java.util.stream.Collectors.*;

List<String> list = s.collect(toList());
Set<String>  set  = s.collect(toSet());
String joined     = s.collect(joining(", "));
Map<Long,User> byId = users.stream()
    .collect(toMap(User::id, u -> u));
```

The collector catalogue

Collector	Produces
toList / toSet	a List or Set
toMap	a Map from key and value functions
joining	a concatenated String
counting	the element count
summingInt / averagingInt	numeric aggregates

toMap and duplicate keys

WATCH OUT

`toMap` throws if two elements map to the same key. When duplicates are possible, supply a merge function — `toMap(k, v, (a, b) -> a)` — to decide which wins.

collect into any shape

collect() with a Collector turns a stream into lists, sets, maps, joined strings, or aggregates. toList/toSet/toMap/joining cover most needs — give toMap a merge function when keys can collide.

MODULE 4 · COLLECTORS & PARALLELISM

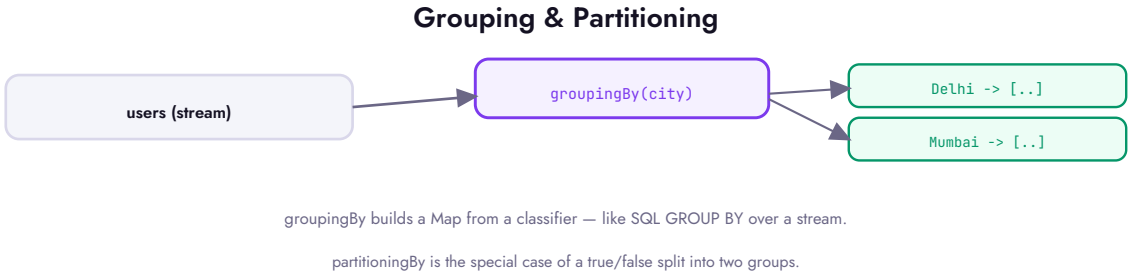
CHAPTER 16

Grouping & Partitioning

Organising stream elements into groups — like SQL GROUP BY.

One of the most powerful collectors is `groupingBy` — it organises elements into a map of groups, the streams equivalent of SQL's `GROUP BY`. `partitioningBy` is its true/false special case.

groupingBy



groupingBy builds a Map from a classifier; partitioningBy splits into true/false groups.

```java
// Map<String, List<User>> keyed by city
Map<String, List<User>> byCity = users.stream()
    .collect(groupingBy(User::city));
```

Downstream collectors

`groupingBy` takes an optional **downstream** collector to process each group — count them, sum a field, or map to another shape — turning grouping into full aggregation.

```java
// count of users per city
Map<String, Long> counts = users.stream()
    .collect(groupingBy(User::city, counting()));

// names per city
Map<String, List<String>> names = users.stream()
    .collect(groupingBy(User::city, mapping(User::name, toList())));
```

partitioningBy

`partitioningBy` splits elements by a predicate into a `Map<Boolean, List<T>>` — the true group and the false group. Use it for a clean two-way split.

```java
Map<Boolean, List<User>> byActive = users.stream()
    .collect(partitioningBy(User::isActive));
```

GROUP BY for streams

groupingBy builds a Map keyed by a classifier, with optional downstream collectors (counting, summing, mapping) to aggregate each group. partitioningBy is the true/false special case for a two-way split.

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CHAPTER 17

Parallel Streams

Processing a stream across multiple cores — and when it actually helps.

A stream can run in **parallel** across multiple CPU cores by changing one method call. It can speed up heavy work — but it is easy to misuse, and often makes things slower.

Going parallel

```java
long count = bigList.parallelStream()
    .filter(this::isExpensiveCheck)
    .count();
// or: stream().parallel()
```

Parallel streams split the work across a shared thread pool (the common ForkJoinPool) and combine the results. For CPU-heavy operations on large datasets, this can use all your cores.

When parallel helps

Good for	Bad for
large datasets	small collections
expensive per-element work	cheap operations
independent operations	ordered or stateful work

The pitfalls

WATCH OUT

Parallel streams have real hazards: shared mutable state causes race conditions, ordering is not guaranteed, and the overhead of splitting can make small or cheap pipelines **slower**. Never assume parallel is faster — measure it.

The rule

Default to sequential streams. Reach for parallel only when you have a large dataset, genuinely expensive independent per-element work, and a benchmark proving it is faster. It is a specialised tool, not a free speed-up.

Parallel: powerful but measured

parallelStream() splits work across cores and can speed up large, CPU-heavy, independent pipelines. But overhead, ordering, and shared state make it a hazard — default to sequential and only go parallel when a benchmark proves it wins.

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CHAPTER 18

Streams Best Practices & Pitfalls

Writing streams that are correct, readable, and appropriate.

Streams are powerful but easy to misuse. A few habits keep them readable and correct — and knowing when not to use a stream is as important as knowing how.

Keep pipelines pure

WATCH OUT

Stream operations should be **stateless** and side-effect-free. Modifying an external list from inside `map` or `filter` breaks in parallel and is hard to reason about — collect a result instead of mutating outside state.

forEach is not a loop replacement

`forEach` is for side effects, not for building results. If you are accumulating into a collection, use `collect` — it is clearer, works in parallel, and expresses intent.

Readability over cleverness

- Break long pipelines across lines, one operation per line.
- Extract complex lambdas into named methods (method references).
- Prefer a simple loop when a stream would be more convoluted.

When not to use a stream

Streams shine for transforming and aggregating data. For simple iteration with side effects, an ordinary `for` loop is often clearer. Do not force a stream where a loop reads better — idiomatic beats clever.

Correct, readable, appropriate

Keep stream pipelines stateless and side-effect-free, use collect (not forEach) to build results, format long pipelines line-by-line, and choose a plain loop when it reads more clearly. Idiomatic beats clever.

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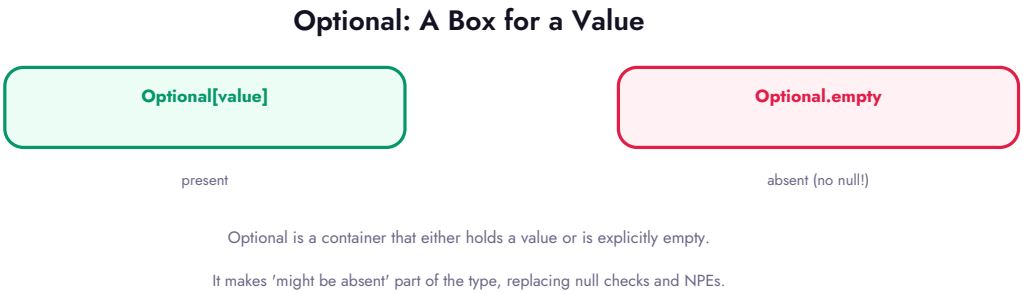
CHAPTER 19

Optional: Taming null

A container type that makes the possibility of absence explicit.

The `NullPointerException` is Java's most infamous bug. `Optional` attacks the root cause by making 'might be absent' part of the type, so the compiler and the reader both see it.

A box for a maybe-value



Optional is a container that either holds a value or is explicitly empty.

```java
Optional<User> found = repository.findById(id);
// found either holds a User or is empty — no null in sight
```

Creating and reading

Method	Does
Optional.of(x)	wrap a non-null value
Optional.ofNullable(x)	wrap a possibly-null value
Optional.empty()	an explicit absence
isPresent / isEmpty	test for a value
get	the value (throws if empty)

Why it beats null

A method returning `Optional<User>` tells the caller, in the type itself, that there may be no user — forcing them to handle the empty case. A method returning `User` that might be null gives no such warning, and the NPE surfaces later, elsewhere.

Absence as a type
Optional is a container that holds a value or is explicitly empty, making possible absence part of the type. Create it with of/ofNullable/empty. It surfaces the missing-value case at compile time, not as a runtime NPE.

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CHAPTER 20

Using Optional Well

The functional methods that make Optional powerful — and how to avoid misusing it.

`Optional` is far more than `isPresent` plus `get`. Its functional methods let you transform and default a value without ever unwrapping it — and using them well is the whole point.

Transform without unwrapping

java

```
String name = repository.findById(id)
    .map(User::name)                // Optional<String>
    .map(String::toUpperCase)
    .orElse("UNKNOWN");             // default if empty
```

The key methods

Method	Does
map	transform the value if present
filter	keep it only if it matches
orElse	a default value if empty
orElseGet	a lazily-computed default
orElseThrow	throw if empty
ifPresent	act on the value if present

orElse vs orElseGet

`orElse(x)` always evaluates `x`, even when a value is present; `orElseGet(() -> x)` computes the default only when needed. Use `orElseGet` when the default is expensive to build.

How not to use Optional

WATCH OUT

Do not call `get()` without checking — it defeats the purpose. Do not use `Optional` for fields, method parameters, or collections. It is designed as a **return type** for 'might not have a result' — nothing more.

Chain, do not unwrap

Use Optional's functional methods — map, filter, orElse/orElseGet, orElseThrow, ifPresent — to handle values without get(). Prefer orElseGet for expensive defaults, and use Optional only as a return type.

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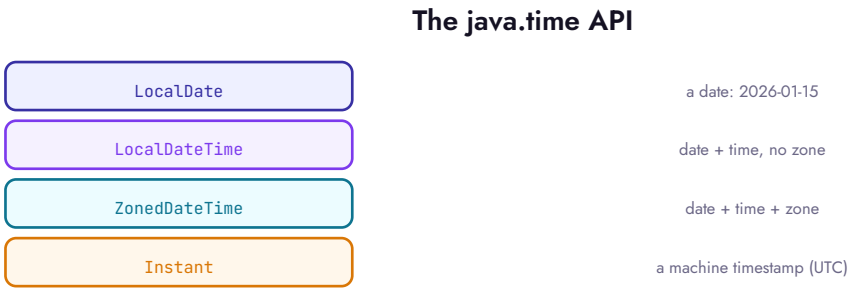
CHAPTER 21

The new Date/Time API

The java.time package that finally fixed dates and times in Java.

Java's old `Date` and `Calendar` were notoriously broken — mutable, confusingly designed, and not thread-safe. Java 8's `java.time` package replaced them with a clean, immutable, well-designed API.

The core types



The java.time types are immutable and thread-safe — replacing the broken Date and Calendar.

The java.time types are immutable and thread-safe — replacing the broken Date and Calendar.

Type	Represents
LocalDate	a date (2026-01-15)
LocalTime	a time of day
LocalDateTime	date and time, no zone
ZonedDateTime	date, time, and zone
Instant	a machine timestamp (UTC)

Everyday use

```java
LocalDate today = LocalDate.now();
LocalDate dob = LocalDate.of(1990, 5, 20);
LocalDate next = today.plusDays(7);       // immutable -> new value
boolean before = dob.isBefore(today);
```

Immutable and fluent

Every `java.time` type is immutable, so operations like `plusDays` return a **new** value rather than mutating the original. This makes them thread-safe and eliminates a whole class of bugs the old API caused.

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Dates done right

java.time replaces the broken Date and Calendar with immutable, thread-safe types: LocalDate, LocalTime, LocalDateTime, ZonedDateTime, and Instant. Operations return new values, making date handling safe and clear.

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CHAPTER 22

Formatting, Durations & Time Zones

Parsing, formatting, measuring, and zoning time correctly.

Beyond the core types, `java.time` handles the harder parts of dates well: formatting and parsing, measuring elapsed time, and the perennial headache of time zones.

Formatting and parsing

```java
DateTimeFormatter fmt = DateTimeFormatter.ofPattern("dd/MM/yyyy");
String text = LocalDate.now().format(fmt);       // "15/01/2026"
LocalDate d = LocalDate.parse("15/01/2026", fmt);
```

`DateTimeFormatter` is immutable and thread-safe — unlike the old `SimpleDateFormat`, which caused subtle bugs when shared across threads. You can safely reuse one formatter everywhere.

Duration and Period

Type	Measures
Duration	time-based amount (hours, seconds)
Period	date-based amount (years, days)

```java
Duration d = Duration.between(start, end);   // e.g. 2h30m
Period age = Period.between(dob, LocalDate.now());
int years = age.getYears();
```

Time zones

`ZonedDateTime` and `ZoneId` handle zones properly, including daylight saving. `Instant` is the zone-neutral point on the timeline — store timestamps as `Instant` (UTC) and convert to a zone only for display.

Format, measure, and zone safely

DateTimeFormatter (thread-safe, unlike SimpleDateFormat) parses and formats; Duration and Period measure time and date amounts; ZonedDateTime and ZoneId handle zones. Store instants in UTC, convert to a zone for display.

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MODULE 6 · MORE & PRACTICE

CHAPTER 23

Other Java 8 Additions

The smaller but genuinely useful improvements beyond streams and lambdas.

Java 8's headline features get the attention, but it also brought many smaller improvements that quietly make everyday code cleaner. A few are worth knowing.

New Map methods

The `Map` interface gained methods that eliminate common boilerplate — the 'get, check null, put' dance becomes a single call.

```java
map.getOrDefault(key, 0);
map.putIfAbsent(key, value);
map.computeIfAbsent(key, k -> new ArrayList<>()).add(item);
map.merge(key, 1, Integer::sum);   // count occurrences
map.forEach((k, v) -> System.out.println(k + "=" + v));
```

String and text helpers

- **String.join** — join elements with a delimiter in one call.
- **StringJoiner** — build delimited strings with prefix and suffix.
- **Base64** — a proper, standard Base64 encoder and decoder at last.

CompletableFuture

For asynchronous work, `CompletableFuture` lets you compose non-blocking operations — run a task, transform its result, combine two futures — far beyond the old `Future`.

```java
CompletableFuture.supplyAsync(() -> fetch(id))
    .thenApply(this::transform)
    .thenAccept(System.out::println);
```

Many quiet improvements

Java 8 also added Map methods (getOrDefault, computeIfAbsent, merge), String.join and StringJoiner, a real Base64 codec, and CompletableFuture for composing async work — small wins that clean up everyday code.

MODULE 6 · MORE & PRACTICE

CHAPTER 24

Best Practices & Functional Style

Habits for writing clean, idiomatic Java 8 code.

Knowing the features is one thing; using them well is another. This closing chapter gathers the habits that make Java 8 code clean, readable, and idiomatic.

Embrace the functional idioms

- **Streams** for transforming and aggregating collections.
- **Method references** where they read more clearly than lambdas.
- **Optional** as a return type for 'might be absent'.
- **java.time** instead of Date and Calendar, always.
- **Comparator helpers** instead of hand-written comparison logic.

But do not overdo it

WATCH OUT

Functional style is a tool, not a religion. A deeply nested stream that no one can read is worse than a clear loop. Favour the version your teammates will understand at a glance.

Prefer immutability

Java 8's style leans functional and immutable — pure lambdas, side-effect-free streams, immutable `java.time` values. Immutable data is easier to reason about and safe to share across threads. Lean that way by default.

Keep learning forward

Everything here is the foundation of modern Java. Later versions (records, sealed classes, the enhanced `toList()`, and more) build directly on these ideas — master Java 8 and the rest of modern Java follows naturally.

Idiomatic, not dogmatic

Use streams, method references, Optional, java.time, and Comparator helpers to write clean functional Java — but favour readability over cleverness, prefer immutability, and remember this is the foundation the rest of modern Java builds on.

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